

Employee Attrition Analysis Dashboard

Descriptive HR Analytics report with Power BI dashboards, insights, and retention recommendations.

1470

Total Employees

237

Employees Left

1233

Active Employees

16.12%

Attrition Rate

Tools & Methods

Power BI | DAX Measures | Data Cleaning | Descriptive Analytics |

HR Analytics

Problem Statement

- The organization is facing increasing employee resignations and needs clarity on turnover patterns.
- HR wants to identify departments and job roles with high attrition.
- The analysis must highlight salary, experience, tenure, demographic, and workplace factors associated with attrition.
- The dashboard should support descriptive HR reporting and retention decisions.

Objective

To identify employee turnover patterns by analyzing department-wise attrition, job role attrition, salary impact, experience-level trends, age and gender patterns, and work-related retention factors using descriptive analytics.

Data Cleaning & Dashboard Preparation

- Imported the employee attrition dataset into PowerBI and checked column consistency.
- Validated categorical fields such as Attrition, Department, Gender, OverTime, JobRole, and JobLevel.
- Ensured numeric fields such as MonthlyIncome, Years At Company, Total Working Years, and Distance From Home were usable for analysis.
- Created grouped columns: Age Group, Salary Range, Experience Group, Tenure Group, and Distance Group.
- Created DAX measures for Total Employees, Employees Left, Active Employees, Attrition Rate, Income Gap, Overtime Attrition Rate, and other KPIs.
- Designed four focused dashboard pages with slicers, KPI cards, charts, heatmaps, and insights.

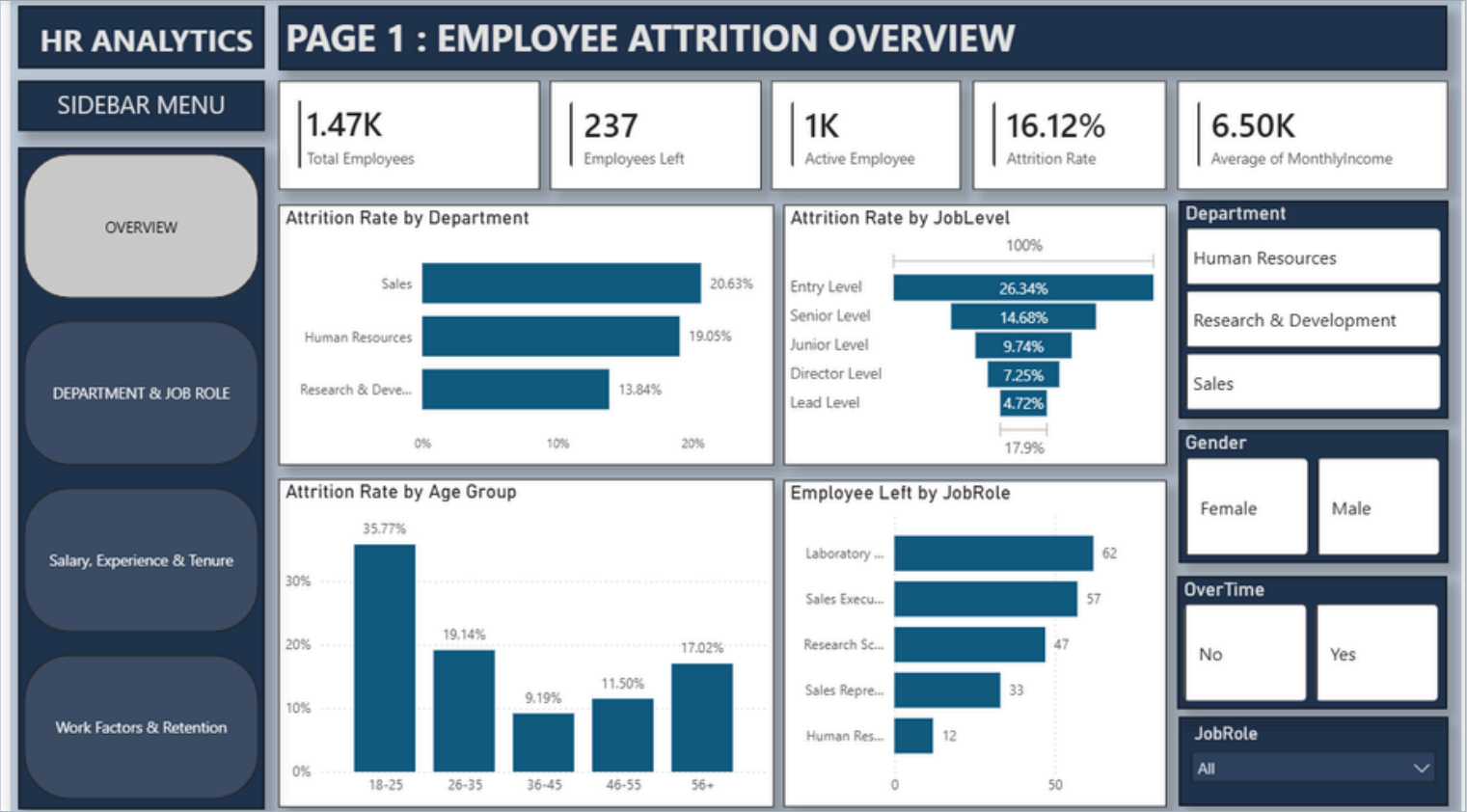
Core DAX Logic

```
Total Employees =  
DISTINCTCOUNT(EmployeeNumber)
```

```
Employees Left =  
CALCULATE([Total Employees],  
    Attrition = "Yes")
```

```
Attrition Rate =  
DIVIDE([Employees Left],  
    [Total Employees], 0)
```

```
Income Gap =  
[Avg Income - Active] -  
[Avg Income - Left]
```

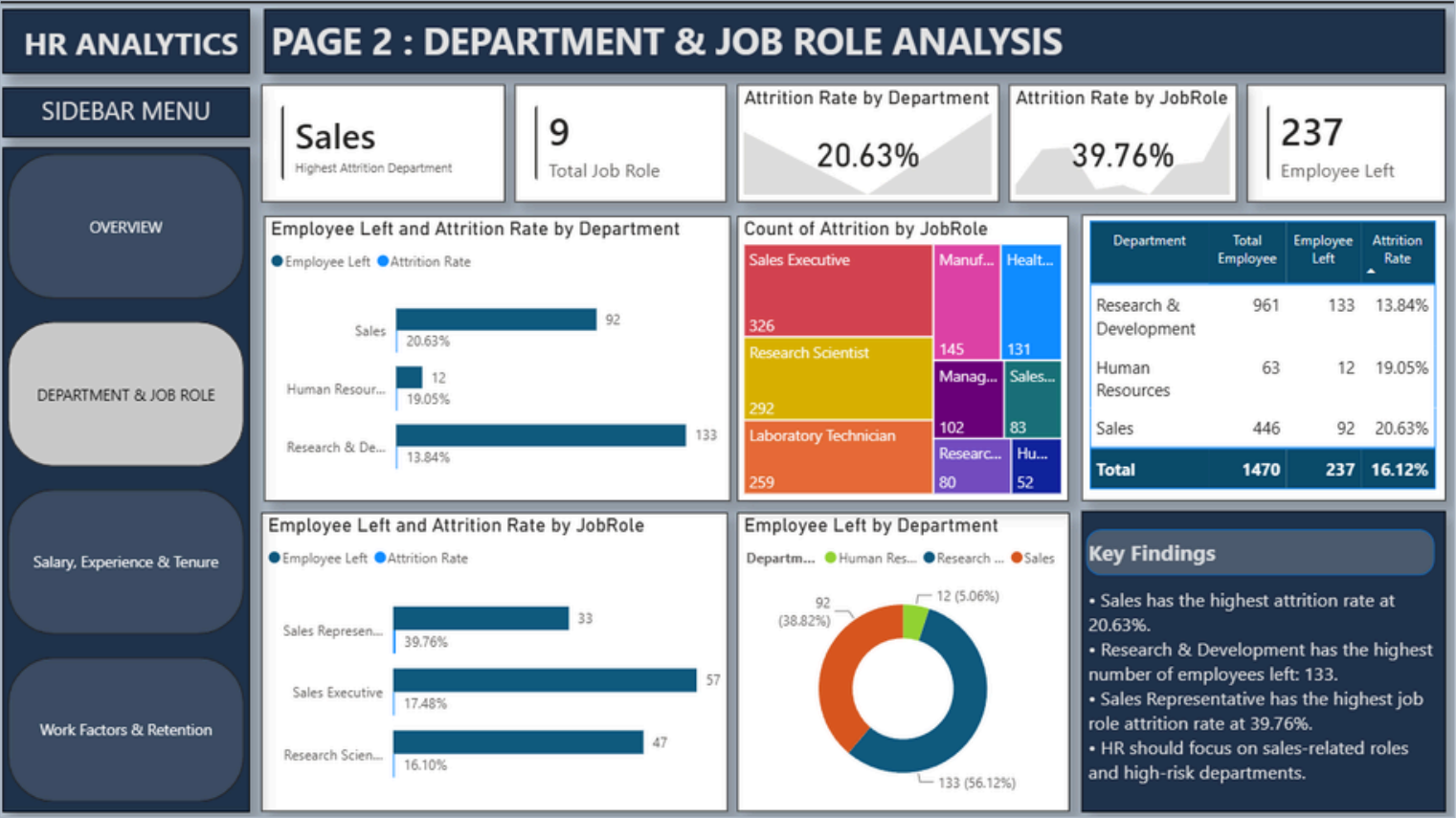


Key Insights

- Overall attrition is 16.12%: 237 employees left out of 1470.
- Sales has the highest department attrition rate at 20.63%; HR follows at 19.05%.
- The (18-25) age group has the highest attrition rate at 35.77%.
- Entry Level employees show the highest job-level attrition risk at 26.34%.
- Job role distribution highlights Laboratory Technician, Sales Executive, Research Scientist, Sales Representative, and HR as important resignation contributors.

Purpose: Provides a high-level view of total workforce, employees left, active employees, overall attrition rate, and broad risk patterns.

Solves: overall attrition clarity

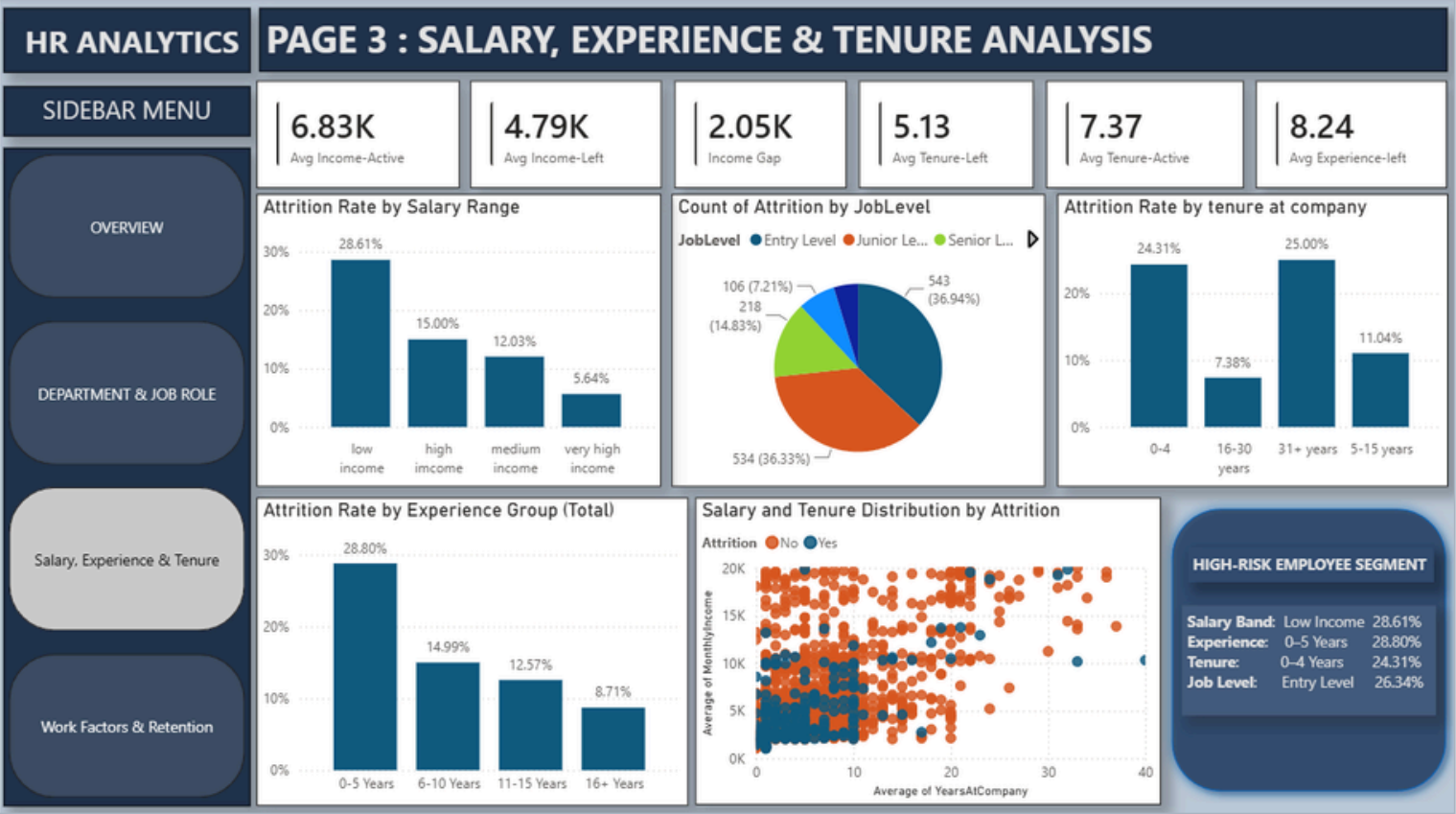


Key Insights

- Sales is the highest-risk department with 20.63% attrition.
- Research & Development has the highest employee-left count at 133 because it has the largest employee base.
- Sales Representative has the highest job role attrition rate at 39.76%.
- Count and rate must be interpreted together -: count shows volume, while rate shows risk.
- HR should prioritize sales-related roles and departments with both high risk and high resignation volume.

Purpose: Identifies the departments and roles where attrition is concentrated, separating resignation volume from attrition risk.

Solves: high attrition departments & roles

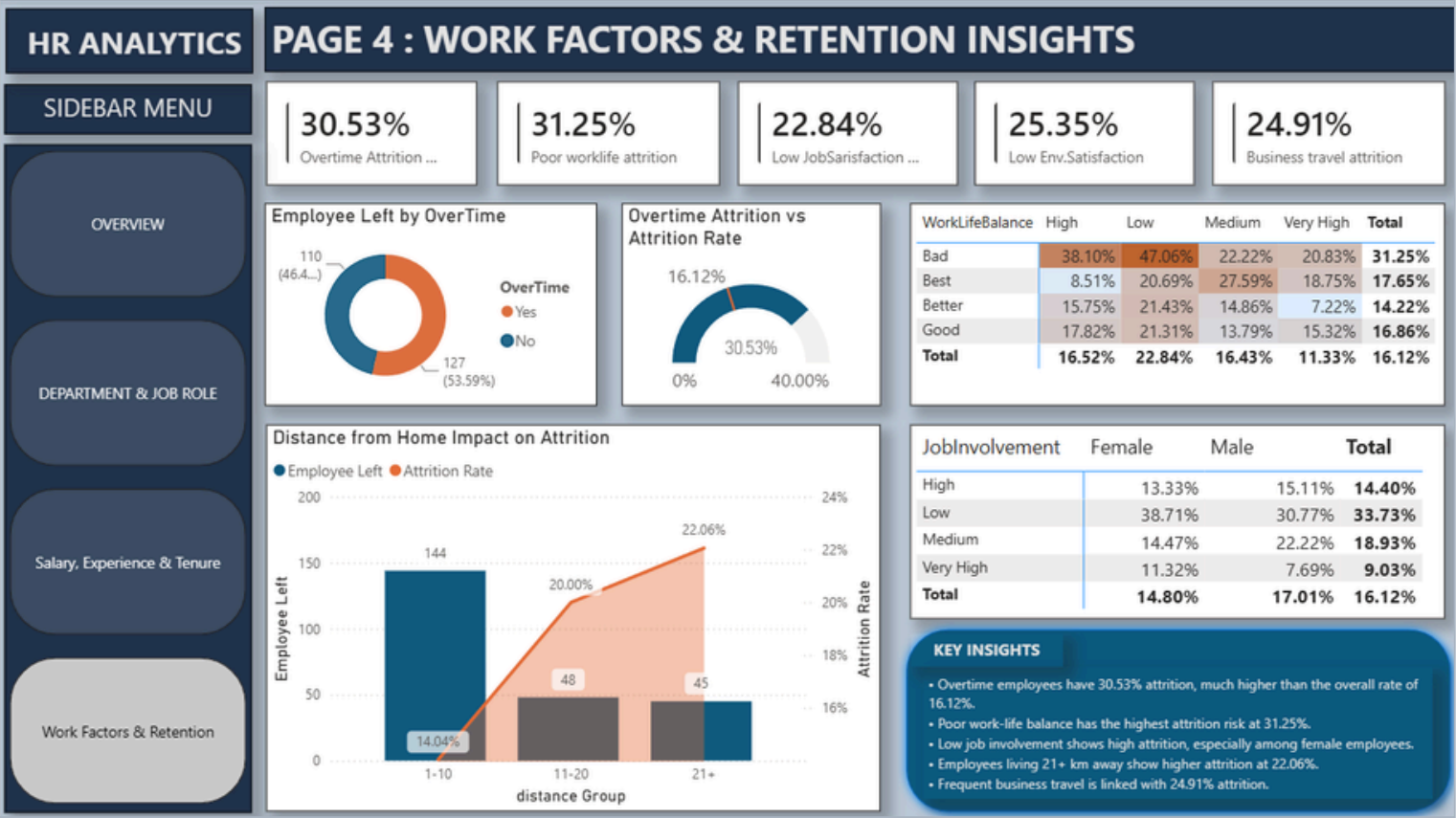


Key Insights

- Low-income employees show the highest salary-based attrition rate at 28.61%.
- Employees who left have lower average income: 4.79K vs 6.83K for active employees.
- The income gap between active and resigned employees is around 2.05K.
- Employees with (0-5) years of experience show 28.80% attrition.
- Employees with (0-4) years of tenure show 24.31% attrition, indicating early-tenure risk.

Purpose: Analyzes whether salary, total experience, and company tenure are associated with higher attrition.

Solves: salary & experience-level trends



Key Insights

- Overtime employees show 30.53% attrition, much higher than the overall rate of 16.12%.
- Poor work-life balance shows 31.25% attrition.
- Low job involvement has the highest work-factor risk at 33.73%.
- Low environment satisfaction is linked with 25.35% attrition.
- Frequent business travel and longer distance from home are also associated with higher attrition.

Purpose: Analyzes work-related factors such as overtime, work-life balance, satisfaction, travel, distance, and job involvement.

Solves: factors associated with resignations

The dashboard pages are aligned with the problem statement. Each page answers a specific HR question and converts it into a practical retention insight.

1. Which departments have high attrition?

- Sales has the highest department attrition rate: 20.63%.
- Research & Development has the highest employee-left count: 133.
- Interpretation: rate shows risk; count shows resignation volume.

2. Which job roles are most affected?

- Sales Representative has the highest job role attrition rate: 39.76%.
- Laboratory Technician and Human Resources also show high attrition risk.
- Action: review sales pressure, incentives, and role support.

3. What experience-level trends are visible?

- 0-5 years total experience has 28.80% attrition.
- 0-4 years company tenure has 24.31% attrition.
- Action: strengthen onboarding, mentoring, and early-career engagement.

4. What factors are associated with resignation?

- Overtime: 30.53%; bad work-life balance: 31.25%.
- Low job involvement: 33.73%; frequent travel: 24.91%.
- Action: reduce workload risk and improve employee engagement.

5. What salary and demographic patterns appear?

- Low-income employees have 28.61% attrition.
- 18-25 age group has 35.77% attrition.
- Entry Level employees show 26.34% attrition risk.

6. What HR can decide from this?

- Prioritize sales-related roles, low-income groups, and early-tenure employees.
- Monitor overtime, work-life balance, job satisfaction, and job involvement.
- Use exit interviews later to verify exact resignation reasons.

Important interpretation: The dashboard identifies patterns associated with attrition, not direct causes. Exact causes would require exit interview responses, resignation reasons, or employee feedback comments.

Recommended HR Actions

- **Salesretention focus:** Review salestargets, incentives, manager support, and career-growth opportunities.
- **Early-tenure support:** Improve onboarding, mentoring, and first-year engagement for employees with 0-4 years at the company.
- **Compensation review:** Reassess salary bands and benefit support for low-income employees.
- **Workload management:** Reduce excessive overtime through staffing review and workload planning.
- **Engagement improvement:** Increase job involvement through role clarity, recognition, participation, and manager check-ins.
- **Flexibility support:** Support frequent travelers and employees living far from the office through flexible work, transport support, or hybrid arrangements.

Limitation of the Analysis

The dataset does not include resignation reasons, exit interview feedback, employee comments, or monthly resignation dates. Therefore, the insights should be presented as associations and patterns, not direct causes.

Conclusion

The dashboard provides a clear descriptive HR analytics report. It highlights where attrition is high, which employee groups need attention, and what retention actions HR can prioritize.



16.12% - Overall
Attrition Rate



39.76% - Highest
Job Role Risk



33.73% - Low Job
Involvement Risk



2.93x
Overtime vs No Overtime Risk